Jane Doe - Curriculum Vitae

Researcher in Statistics

Statistical methods, Machine learning

Sparse models · Multi-omics integration · Reproducible research

Born January 1, 1990
Example citizen

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https://example.org
https://github.com/example

Professional address

Example Institute
1 Example Street
00000 Example City

Brief summary of activities

Activity	since 2010
Production	2 papers, 2 talks
Students	1 PhD, 1 Master
Teaching	Statistics in Action with R, Convex Optimization

Experience

2011–	Researcher, Example Institute
2008–2011	Postdoc, Example University
supervisor	Dr. Example

Education

2015	Habilitation in Mathematics
Title	*Sparse methods for complex data*
2003	PhD in Applied Mathematics

Scientific activities

Grants

On going Projects

2023–2027	DISCERN -- Discovering the causes of a poorly understood cancer
Partners	20 partners
Description	A five-year consortium project combining statistical genomics, imaging, and epidemiological cohort data to identify early biomarkers of a poorly understood cancer subtype. My group leads the statistical modelling work package, developing sparse graphical models for multi-omics integration across the twenty partner sites.
Support	Horizon Europe

Past

2019–2023	EcoNet -- Advanced statistical modelling of ecological networks
Support	French National Research Agency (ANR)
2017–2019	SEARS -- Sampling strategies and network analysis
Leader	Mathieu Thomas

Other grants

2010–2012	Older grant, listed only in the long version
Support	French National Research Agency (ANR)

Activities

Scientific committees

	Councils
since 2022	Member of the Foundation Council
2020–2024	PhD Committee (full list)
role	Reviewer for 12 PhD theses

Students

PhD students

2021–2024	Jane Roe -- Sparse statistical methods for large-scale multi-omics data
Co-supervisor	Prof. Example
Description	Developed sparse graphical model estimators for integrating heterogeneous omics data sources, with applications to the DISCERN consortium. Defended in 2024; now a postdoctoral researcher at Example University.

Master students

2023	John Roe -- Poisson-lognormal models for count data
Co-supervisor	Dr. Example

Teaching

Approximately 2000 hours of teaching. Lecturer at Example University from 2015 to 2024.

2020–24	Statistics in Action with R
Msc	Probabilistic models, data analysis, R programming
2017	Convex Optimization (12h course)
Msc	Gradient methods, Newton method, Proximal methods

Scientific productions

Publications

Selected Publications

[JP1] J. Doe and J. Roe. "A Toy Example for the quarto-cv-blocks Extension". In: *Journal of Reproducible Examples* 1 (2023), pp. 1–10. DOI: 10.0000/example.2023.001.

[JP2] J. Doe. "Demonstrating a Fictitious Conference Paper". In: *Proceedings of the Example Conference*. 2021, pp. 42–50.

Selected Talks

[CT1] J. Doe. "Reproducible CVs with Quarto". In: *Example Workshop on Reproducible Documents*. Example City, 2022.

[CT2] J. Doe and J. Roe. "A Fictitious Talk for the quarto-cv-blocks Example". In: *Proceedings of the Other Example Conference*. 2020, pp. 12–20.

DISCERN

As part of the DISCERN project, my group develops sparse graphical model estimators for integrating multi-omics data. Given a sample covariance matrix S, we estimate a sparse precision matrix $\hat{\Theta}_\lambda$ by solving

$$\hat{\Theta}_\lambda = \arg\min_{\Theta \succ 0} \; -\log\det\Theta + \mathrm{tr}(S\Theta) + \lambda\|\Theta\|_1.$$

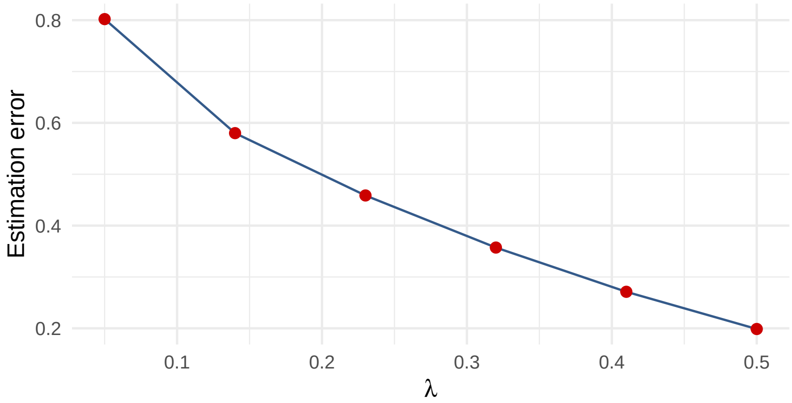

Figure 1: Estimation error as a function of the sparsity penalty

Increasing the penalty λ trades estimation error for a sparser, more interpretable graph, see Figure 1.

Another project

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See Table 1.

Lambda	Error	Sparsity
0.05	0.79	12%
0.20	0.42	48%
0.50	0.19	81%

Table 1: Estimation error and graph sparsity for a few penalty values